

# Data Analysis Workbook

## Step-by-step guidance for Workflow B

|                    |                                   |
|--------------------|-----------------------------------|
| Client             | Generation Bridge Foundation      |
| Document reference | MOM-2026-WSB-02                   |
| Classification     | Participant material - Workflow B |
| Version            | 3.0   July 2026                   |

This workbook explains how to complete each analysis task in Excel. Package results as Findings Memos when answering Workflow A's requests. You do not need to write code. If you use Python or R, the same documentation standards apply.

## Concepts in plain English

| Term           | What it means                                                         |
|----------------|-----------------------------------------------------------------------|
| Pivot table    | A table that groups and summarises your data automatically            |
| XLOOKUP        | Finds a value in one table and returns a matching value from another  |
| Placement rate | Placed completers divided by total completers - define both parts     |
| Correlation    | Two things move together; it does not prove one causes the other      |
| Confounder     | A hidden factor that affects both things you are comparing            |
| Selection bias | Those who attend more may already differ in ways that affect outcomes |

## From request to Findings Memo

1. Read A's request. Restate the decision question in your own words.
2. Decide which tables and fields answer it. Clean only what you need first.
3. Compute the result. Note definitions and sample limitations.
4. Write the Findings Memo in plain language. Attach at most two curated charts.
5. Update the Request Log status to 'Returned'.

## Worked approach to DB-3 (attendance and outcomes)

1. In programme\_attendance, create a helper column that is 1 when attended and 0 when not, normalising mixed Y/N/1/0 coding.
2. Build a pivot for each young person's attendance rate.
3. Use XLOOKUP to bring in placement from employment\_outcomes.
4. Compare average attendance of those placed with those not placed.
5. Interpret the gap in two or three sentences and name confounders (for example transport and motivation).

## Reporting standards

- State the placement definition you used and why.
- Never delete an outlier silently - investigate and record the decision.
- Give every chart a title that states the finding.
- Note the sample-size limitation versus the 2,847 in D-01.
- Never attach the raw workbook to a Findings Memo.